

Event-gated delayed feedback and precision–dissipation trade-offs in autonomous oscillator clocks

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Autonomous clocks built on an oscillatory degree of freedom can exceed precision–dissipation bounds derived for classical Markov-jump or overdamped dynamics, such as the thermodynamic uncertainty relation (TUR). Timekeeping in such devices requires not only a dissipative oscillator but also an *extraction channel* that converts the continuous phase into discrete, countable ticks. We introduce a minimal, coarse-grained description of an *event-gated delayed-feedback* oscillator clock, in which the escapement is realized as an event-triggered reinjection pulse delivered a fixed latency τ_d after a detected threshold crossing, and the ticks are read out through a finite-bandwidth detector with intrinsic dead time. Using phase reduction we obtain closed-form expressions for the resolution, accuracy and Allan variance. We also give a minimal numerical benchmark and a reproducible simulation protocol that separates engine noise, extraction bandwidth and delay feedback. In the collective limit of M independent escapement units we show analytically, via a central-limit argument consistent with existing mean-field numerics, that the phase diffusion scales as $D_\phi(M) \propto M^{-1}$, so that the accuracy grows as $\mathcal{N} \propto M$ while the engine entropy production grows as $\dot{\Sigma} \propto M$; the TUR ratio therefore approaches a finite, M -independent constant. This reproduces and extends—beyond the small particle numbers accessible to direct simulation—the persistence of TUR violation across the quantum-to-classical transition reported for optomechanical pendulum clocks. Finally we formulate, as a falsifiable hypothesis rather than a theorem, a precision–dissipation balance in which the extraction channel contributes its own entropy production, and we give measurable discriminators, including delay-dependent suppression of phase diffusion. The framework unifies stochastic-transition clocks, optomechanical pendulum clocks and delay-feedback clocks within a common thermodynamic accounting and requires no assumptions specific to the quantum regime.

I. INTRODUCTION

A clock is any system whose irreversible dynamics produce a countable, quasi-periodic sequence of events. Thermodynamic approaches to timekeeping have made this precise by bounding the achievable accuracy in terms of dissipation [1–6]. A central benchmark is the thermodynamic uncertainty relation (TUR) [7–9], which for a clock takes the form

$$\mathcal{N} \leq \frac{1}{2} \langle \tau \rangle \dot{\Sigma}, \quad (1)$$

where $\mathcal{N}$ is the accuracy, $\langle \tau \rangle$ the mean inter-tick time and $\dot{\Sigma}$ the entropy-production rate. The bound (1) is rigorous for systems obeying Markovian rate equations or overdamped Langevin dynamics, but most practical clocks are based on an underdamped oscillatory degree of freedom and are not constrained by it [10–14]. Violations of (1) have accordingly been reported in classical pendulum clocks [10, 11] and, recently, in an optomechanical pendulum clock operated from the quantum to the classical regime [14].

Two features of these oscillator clocks deserve closer scrutiny. First, the reported TUR violation is associated specifically with *filtered* ticks: the raw emission record is post-processed through a finite-bandwidth filter that suppresses spurious short-time events, and the entropy cost of that data processing has been identified as an open problem [14, 15]. Treating the extraction stage as a free, time-independent operation is therefore an approximation whose validity controls whether a given violation

is physical or an artifact of accounting. Second, in the many-unit limit, accuracy and dissipation are both expected to grow linearly with the number of escapement elements [14], so that the violation persists into the classical limit; but this expectation has so far been supported only by direct simulation up to small particle numbers, beyond which the state-space cost becomes prohibitive and analytic methods are required.

Here we address both points with a single, deliberately minimal model. We describe the escapement not as an instantaneous nonlinearity but as an *event-triggered delayed-feedback* channel: a detected threshold crossing schedules a reinjection pulse after a fixed latency τ_d , closing a delay loop with the oscillator. Delayed feedback in circulating nonlinear systems is a well-studied route to discrete operating branches and mode selection [22, 23], and the event-triggered form is the natural classical counterpart of an escapement that “locks” and “releases” once per resonance. The ticks are then read out through a finite-bandwidth detector with a built-in dead time, modeled as in the optomechanical analysis [14, 18].

Within this framework we (i) reduce several known clock models to common limits; (ii) derive, via phase reduction, closed-form expressions for the resolution, accuracy and Allan variance; (iii) establish analytically that $D_\phi(M) \propto M^{-1}$ in the collective limit, yielding $\mathcal{N} \propto M$, $\dot{\Sigma} \propto M$ and an M -independent TUR ratio; and (iv) formulate a testable precision–dissipation balance that includes the extraction channel, together with measurable discriminators. No assumption specific to the quantum

regime enters; the quantum optomechanical clock [14] is recovered as one realization.

II. EVENT-GATED DELAYED-FEEDBACK CLOCK

A. Oscillator and escapement

The timekeeping element is a damped harmonic oscillator with natural frequency ω and damping γ , written in quadratures (x, p) . Energy lost to damping is replenished by discrete reinjection pulses delivered by the escapement,

$$\dot{x} = \omega p - \frac{\gamma}{2} x, \quad (2)$$

$$\dot{p} = -\omega x - \frac{\gamma}{2} p + A \sum_k \delta(t - t_k), \quad (3)$$

where A is the impulse strength and $\{t_k\}$ are the firing times of the escapement. Equations (2)–(3) are the linear envelope of any underdamped limit-cycle clock; the nonlinearity and irreversibility reside in how $\{t_k\}$ are generated.

The escapement is gated by the oscillator phase. We model the release of a candidate event by an inhomogeneous point process with phase-dependent intensity

$$\lambda(\phi) = \lambda_0 \exp\left[-\frac{\Delta(\phi)^2}{2\sigma_\Delta^2}\right], \quad \Delta(\phi) = \Delta_0 - \chi x(\phi), \quad (4)$$

so that events are overwhelmingly likely only when the oscillator passes through a narrow resonance window ($\Delta \simeq 0$), of width set by σ_Δ and sensitivity χ . Equation (4) is the classical, coarse-grained counterpart of a resonance-conditioned escapement: in the optomechanical realization the gate is the displacement-tuned cavity-emitter detuning, and the released event is a single intra-cavity photon [14].

B. Event-triggered delayed reinjection

A released candidate event at time s_j does not act on the oscillator instantaneously. Detection and actuation incur a fixed latency τ_d , so the reinjection pulse fires at

$$t_j = s_j + \tau_d. \quad (5)$$

Equation (5) promotes the device from a memoryless escapement to a genuine delayed-feedback loop: the kick applied at time t_j depends on the oscillator state at the earlier time $s_j = t_j - \tau_d$. In the deterministic (noise-free) limit, Eqs. (2)–(5) define a delay dynamical system with two distinct dimensionless control parameters. The event-count delay number $K_{\text{ev}} = \lambda_0 \tau_d$ is the mean number of candidate release attempts during one delay interval. The linear gain-delay product $K_{\text{lin}} = \kappa \tau_d$ controls small-signal stability of the locked phase. Only K_{lin}

is the conventional gain-delay product of linear delayed-feedback theory; K_{ev} is a gated-event occupancy measure. We work throughout in the stable-locking regime, where a unique attracting limit cycle exists and the clock ticks regularly.

C. Finite-bandwidth tick extraction

The countable output of the clock is obtained not from the candidate reinjection times $\{t_j\}$ directly but from a detector with finite bandwidth Γ , threshold I^* , pulse response amplitudes q_j , and optional comparator blanking time τ_{ref} . We model the analog detector current as the leaky-integrator process

$$\dot{I}(t) = -\Gamma I(t) + \sum_j q_j \delta(t - t_j), \quad \Gamma > 0, \quad (6)$$

with solution

$$I(t) = \sum_{j:t_j < t} q_j e^{-\Gamma(t-t_j)}. \quad (7)$$

A detected tick time T_n is the next upward threshold crossing after the previous blanking interval has elapsed,

$$T_{n+1} = \inf \left\{ t > T_n + \tau_{\text{ref}} : I(t^-) < I^* \leq I(t^+) \right\}. \quad (8)$$

Equations (6)–(8) distinguish three physical operations that are often conflated: candidate event production, analog integration, and digital/comparator thresholding. The finite relaxation time Γ^{-1} merges closely spaced candidate events into one detectable excursion, while τ_{ref} represents a true dead time. In the limit $q_j > I^*$ and $\tau_{\text{ref}} \rightarrow 0$ almost every candidate event is counted; in the opposite limit, several candidate events must arrive within a window $\sim \Gamma^{-1}$ to produce one counted tick. The analytic results in Secs. IV–V are derived for $\tau_{\text{ref}} = 0$; a nonzero blanking time adds a lower bound on the inter-tick interval $\langle \tau \rangle \geq \tau_{\text{ref}}$ and reduces the effective resolution, but does not alter the scaling laws. The key conceptual point of this work is that the extraction stage is not free post-processing: it is a physical dynamical subsystem and carries its own dissipation, quantified in Sec. VI.

III. REDUCTION TO KNOWN CLOCK MODELS

The model of Sec. II contains several established clocks as limits, which fixes its scope and its relation to prior bounds.

- **Stochastic-transition (Markovian) clock.** For $\gamma \gg \omega$ the oscillator is overdamped and the coarse-grained clock variable reduces to a renewal-like biased random walk; the gate (4) becomes effectively

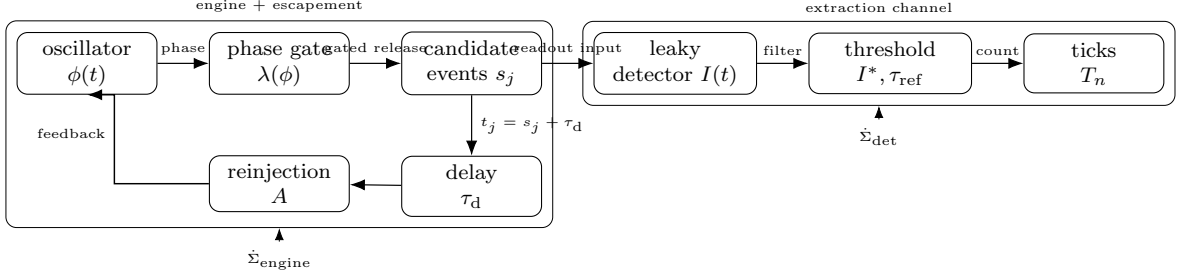


FIG. 1. Architecture of the event-gated delayed-feedback clock. The oscillator and escapement form the engine: the phase gate produces candidate events s_j , which are reinjected only after the latency τ_d . The tick record is produced by a separate extraction channel: a leaky detector followed by a threshold/dead-time stage. The entropy accounting below separates engine dissipation, $\dot{\Sigma}_{\text{engine}}$, from detector dissipation, $\dot{\Sigma}_{\text{det}}$.

state-independent and Eqs. (2)–(8) collapse to a jump process. The bound (1) then applies in its usual inequality form, with saturation only in special tightly coupled or unicyclic limits [1, 2, 5].

- **Optomechanical pendulum clock.** For $\gamma \ll \omega$, $\tau_d \rightarrow 0$, and a single resonance-gated emitter event per half-period, the model is the semiclassical envelope of the optomechanical pendulum clock of Ref. [14], with $I(t)$ the photodetector low-pass filter.
- **Delay-feedback clock.** For finite latency and appreciable loop gain, Eq. (5) produces a genuine delay-feedback oscillator clock. When the delay is of order the oscillation period, $\tau_d \sim \omega^{-1}$, the relevant regime is resonant delayed feedback or entrainment. The long-delay spatio-temporal correspondence of Refs. [22, 23] instead requires a separation of scales $\tau_d \gg \omega^{-1}$ together with a delay large compared with the local relaxation time of the loop.

A clock that violates (1) in one limit and obeys it in another shows that the violation is a property of the underdamped oscillatory sector, not of the extraction stage alone; this motivates treating the two contributions to dissipation separately (Sec. VI).

IV. PHASE REDUCTION: RESOLUTION, ACCURACY, ALLAN VARIANCE

In the stable-locking regime the slow dynamics collapse onto the limit cycle and are captured by a single phase variable obeying

$$\dot{\phi} = \omega + \xi_{\phi}(t), \quad \langle \xi_{\phi}(t) \xi_{\phi}(t') \rangle = 2 D_{\phi} \delta(t - t'), \quad (9)$$

where D_{ϕ} (units s^{-1} , i.e. $\text{rad}^2 \text{s}^{-1}$) is the effective phase-diffusion coefficient that aggregates thermal, shot, and detector noise projected onto the cycle. A detected tick

is emitted each time the phase advances by a fixed increment $\Delta\phi$ ($\Delta\phi = 2\pi$ for one tick per period, $\Delta\phi = \pi$ for the tick-tock pattern of a pendulum clock).

Resolution. The mean inter-tick interval is the mean first-passage time of (9) across $\Delta\phi$, giving

$$\langle \tau \rangle = \frac{\Delta\phi}{\omega}, \quad \nu \equiv \frac{1}{\langle \tau \rangle} = \frac{\omega}{\Delta\phi}. \quad (10)$$

Accuracy. In the stable-locking regime the drift dominates the diffusion, i.e. $D_{\phi} \ll \omega \Delta\phi$ (equivalently, $\mathcal{N} \gg 1$); this holds for any reasonable clock design in the regime of interest. Under this condition the first-passage-time variance across $\Delta\phi$ of a drift-diffusion process is $\text{Var}(\tau) = 2 D_{\phi} \Delta\phi / \omega^3$, so that

$$\mathcal{N} \equiv \frac{\langle \tau \rangle^2}{\text{Var}(\tau)} = \frac{\Delta\phi \omega}{2 D_{\phi}}. \quad (11)$$

Equation (11) is dimensionless, as required. For a clock producing one tick per period ($\Delta\phi = 2\pi$), it gives $\mathcal{N} = \pi\omega/D_{\phi}$.

Allan variance. For uncorrelated successive intervals the overlapping Allan variance follows the standard white-frequency form [19]

$$\sigma_A^2(m) = \frac{1}{m \mathcal{N}}, \quad (12)$$

the agreement of which with single-trajectory estimates at small averaging factor m is a direct test of the renewal assumption underlying (11).

Role of the delay. Each delayed kick $A\delta(t - t_j)$ projects onto the phase through the infinitesimal phase-response curve (PRC) $Z(\phi)$ [20, 21]. For a phase-gated process, the mean frequency shift and the linear restoring response are different projections. If the gate is narrow and centered at phase ϕ_g , the mean phase increment per event scales as $AZ(\phi_g)$, whereas the restoring response to a small phase error depends on the local slope $AZ'(\phi_g)$. Linearization about the locked cycle therefore gives

$$\kappa \simeq \bar{\lambda} A Z'(\phi_g), \quad \bar{\lambda} = \frac{1}{2\pi} \int_0^{2\pi} \lambda(\phi) d\phi, \quad (13)$$

up to pulse-shape factors of order unity. For $Z(\phi) = Z_0 \sin \phi$, this becomes $\kappa \simeq \lambda A Z_0 \cos \phi_g$; a gate centered near $\phi_g = 0$ has vanishing mean phase shift but maximal restoring slope. This resolves the apparent conflict between a narrow gate and a sinusoidal PRC.

The reference coefficient $D_\phi^{(0)}$ used below is defined on the open-loop reference cycle: the delay feedback is set to $\kappa = 0$, while an auxiliary instantaneous drive keeps the mean amplitude and frequency equal to those of the locked delayed cycle. The PRC and noise projection are evaluated on this reference cycle. The delay then enters only through the linear response of the locked cycle to a pulse applied at the delayed phase. Writing the phase fluctuation as $\delta\phi(t)$ and denoting the small-signal delayed loop kernel by $K_{\tau_d}(s)$, the required scalar linearization is

$$\dot{\delta\phi}(t) = - \int_0^\infty K_{\tau_d}(s) [\delta\phi(t) - \delta\phi(t-s)] ds + \xi_\phi(t), \quad (14)$$

where the kernel contains the phase-response curve, the pulse shape, and the detection latency. Throughout, we use the one-sided (engineering) Fourier convention $\hat{K}(\Omega) = \int_0^\infty K(s) e^{-i\Omega s} ds$, consistent with $\partial_t \rightarrow i\Omega$. In the frequency domain Eq. (14) then gives

$$\widetilde{\delta\phi}(\Omega) = \frac{\widetilde{\xi}_\phi(\Omega)}{i\Omega + \hat{K}_{\tau_d}(0) - \hat{K}_{\tau_d}(\Omega)}. \quad (15)$$

The long-time diffusion is therefore controlled by the zero-frequency slope of the feedback denominator,

$$D_\phi^{\text{eff}} = \frac{D_\phi^{(0)}}{|1 + \Lambda(\tau_d)|^2}, \quad (16)$$

$$\Lambda(\tau_d) := -i \partial_\Omega [\hat{K}_{\tau_d}(0) - \hat{K}_{\tau_d}(\Omega)]_{\Omega=0}.$$

The sign is fixed by requiring $D_\phi^{\text{eff}} < D_\phi^{(0)}$ for stabilizing ($\kappa > 0$) feedback: with the engineering convention $\hat{K}(\Omega) = \kappa e^{-i\Omega\tau_d}$, one obtains $\partial_\Omega [\hat{K}(0) - \hat{K}(\Omega)]_0 = i\kappa\tau_d$, so $\Lambda = -i \cdot i\kappa\tau_d = +\kappa\tau_d > 0$, yielding $|1 + \Lambda|^2 = (1 + \kappa\tau_d)^2 > 1$ as required. For a pure delayed proportional feedback term, $-\kappa[\delta\phi(t) - \delta\phi(t - \tau_d)]$, Eq. (16) reduces to the low-frequency expression

$$D_\phi^{\text{eff}} = \frac{D_\phi^{(0)}}{(1 + \kappa\tau_d)^2}, \quad 1 + \kappa\tau_d > 0. \quad (17)$$

Equation (17) applies to pure delayed phase stabilization with frequency-independent small-signal gain. Periodic dependence on $\omega\tau_d$ is a separate, resonantly gated case: it appears when the gain is evaluated on a periodically sampled oscillator. A minimal sinusoidal phase-response model gives

$$\Lambda(\tau_d) \simeq \kappa\tau_d \mathcal{F}(\omega\tau_d), \quad \mathcal{F}(z) = \cos(z + \psi) \hat{g}(\omega), \quad (18)$$

where ψ is the phase-response-curve offset, and $\hat{g}(\omega)$ is the Fourier amplitude of the reinjection pulse shape $q(t)$

evaluated at the fundamental oscillation frequency ω (not the analysis frequency Ω). A sinc factor is recovered only for particular finite-width pulse shapes. The robust prediction is therefore not a specific sinc law but a delay-dependent modulation of D_ϕ^{eff} , with extrema determined by the experimentally measured loop transfer function.

V. COLLECTIVE LIMIT AND THE PERSISTENCE OF TUR VIOLATION

We now treat M identical, independent escapement units sharing the same oscillator, the classical-limit configuration in which fluctuations are expected to vanish. Each unit fires its own resonance-gated events; the oscillator is driven by their sum.

Drift. In the linear-damping, weak-saturation regime the aggregate reinjection rate is the sum of M independent intensities, so the mean kick per cycle, and hence the limit-cycle action J , scale as

$$J(M) \propto M. \quad (19)$$

Phase diffusion. The per-cycle phase kick is an *average* over the M units. If each unit contributes an independent stochastic phase impulse of variance v_1 per unit time, the variance of the normalized aggregate impulse is suppressed by the central limit theorem,

$$D_\phi(M) = \frac{D_1}{M}, \quad (20)$$

where D_1 is the single-unit value. Equation (20) is the analytic statement of the $1/\sqrt{M}$ suppression of the conditional noise terms found in the mean-field treatment of the multi-emitter optomechanical clock [14, 17]: there the stochastic increments enter as dN/M while the number of jumps grows as M , giving a net noise amplitude $\propto M^{-1/2}$ and hence a phase variance $\propto M^{-1}$. The scaling (20) assumes that the M units are statistically independent. Coherent inter-unit effects—such as superradiant enhancement, which is listed as a future direction in Ref. [14]—would modify both the prefactor and the exponent; their onset is observable as a deviation from the linear $\mathcal{N}(M)/M = \text{const}$ test of Sec. VIII.

Consequences. Substituting (20) into (11),

$$\mathcal{N}(M) = \frac{\Delta\phi\omega}{2D_1} M \equiv \mathcal{N}_1 M \propto M. \quad (21)$$

Because the M units dissipate independently, the engine entropy production is extensive,

$$\dot{\Sigma}_{\text{engine}}(M) = M \dot{\Sigma}_1 \propto M. \quad (22)$$

The TUR ratio is therefore *M-independent*,

$$\mathcal{R}(M) \equiv \frac{2\mathcal{N}(M)}{\langle \tau \rangle \dot{\Sigma}_{\text{engine}}(M)} = \frac{2\mathcal{N}_1}{\langle \tau \rangle \dot{\Sigma}_1} = \mathcal{R}(1), \quad (23)$$

so that if a single underdamped unit violates (1) ($\mathcal{R} > 1$), the violation *persists unchanged* into the macroscopic limit $M \rightarrow \infty$, where the clock becomes deterministic. This equality is *exact*, not asymptotic: it follows directly from the simultaneous linearity of both $\mathcal{N}(M)$ and $\dot{\Sigma}_{\text{engine}}(M)$ in M under the independence assumption. Equation (23) reproduces analytically the persistence of TUR violation across the quantum-to-classical transition reported numerically up to small M in Ref. [14], and extends it to arbitrary M , which direct simulation cannot reach. It also makes explicit that the macroscopic violation is not in tension with the classical TUR derivation: the latter assumes Markovian/overdamped dynamics, a condition violated by the underdamped oscillatory sector at every M .

VI. THERMODYNAMICS OF TICK EXTRACTION

The accuracy (11) refers to the *detected* ticks of Eqs. (6)–(8), whose definition involves the extraction channel. We therefore decompose the total entropy production into three physically distinct contributions,

$$\dot{\Sigma}_{\text{total}} = \dot{\Sigma}_{\text{osc}} + \dot{\Sigma}_{\text{esc}} + \dot{\Sigma}_{\text{det}}, \quad (24)$$

namely (i) dissipation that sustains the oscillation against damping, (ii) dissipation in the event-gated escapement that converts continuous motion into candidate events, and (iii) dissipation in the finite-bandwidth detector that converts candidate events into a counted output. In the small-escapement clocks of interest, the dominant contribution is the escapement: the energy needed to *produce* the ticks far exceeds that needed to *maintain* the oscillation, the opposite of a macroscopic friction-dominated pendulum [14].

A widely used clock TUR [12, 16] bounds the accuracy by the engine entropy production,

$$\frac{2\mathcal{N}}{\langle \tau \rangle \dot{\Sigma}_{\text{engine}}} > 1 \quad (\text{observed violation}), \quad (25)$$

where $\dot{\Sigma}_{\text{engine}} = \dot{\Sigma}_{\text{osc}} + \dot{\Sigma}_{\text{esc}}$. We propose, as a falsifiable hypothesis rather than a proven inequality, that the violation may be reduced or eliminated once the dissipative cost of the extraction channel is included:

$$\frac{2\mathcal{N}}{\langle \tau \rangle (\dot{\Sigma}_{\text{engine}} + \dot{\Sigma}_{\text{det}})} \stackrel{?}{\leq} 1. \quad (26)$$

We stress that (26) is *not* guaranteed: the extraction channel may be cheap enough that the inequality still fails, and indeed the entropic cost of converting microscopic ticks into a classical signal has been shown to be finite but nonzero [15]. The contribution of the present framework is to make $\dot{\Sigma}_{\text{det}}$ an explicit dynamical quantity—the dissipation of the leaky integrator and

comparator defined by Eqs. (6)–(8)—so that the balance (26) can be measured rather than assumed, and to identify the parameter regimes (detector bandwidth Γ , threshold I^* , dead time) in which the extraction cost is non-negligible. This sharpens, and is complementary to, the open problem stated in Refs. [14, 15].

A full stochastic-thermodynamic computation of $\dot{\Sigma}_{\text{det}}$ is beyond the scope of the present coarse-grained treatment, but the leaky integrator nevertheless admits a useful order-of-magnitude estimate. If the analog detector current is modeled as a dissipative coordinate with effective detector-noise strength D_I , then the housekeeping entropy production rate of the analog stage scales as

$$\frac{\dot{\Sigma}_{\text{det}}}{k_B} \sim \frac{\Gamma}{D_I} \langle (I - \langle I \rangle)^2 \rangle, \quad (27)$$

up to a device-dependent prefactor of order unity. For Poisson-distributed candidate events of mean rate $\bar{\lambda}$ and identical pulse amplitude q , Eqs. (6)–(7) give the stationary moments

$$\langle I \rangle = \frac{q\bar{\lambda}}{\Gamma}, \quad \text{Var}(I) = \frac{q^2\bar{\lambda}}{2\Gamma}, \quad (28)$$

so that

$$\langle I^2 \rangle = \frac{q^2\bar{\lambda}^2}{\Gamma^2} + \frac{q^2\bar{\lambda}}{2\Gamma}. \quad (29)$$

Equations (27)–(29) are sufficient for the intended falsifiability test: they show explicitly that $\dot{\Sigma}_{\text{det}}$ increases with detector bandwidth, pulse size, and candidate-event rate, and they provide a concrete scale against which $\dot{\Sigma}_{\text{engine}}$ can be compared in Eq. (26).

VII. MINIMAL NUMERICAL BENCHMARK

The analytic claims above are deliberately separable from microscopic implementation. A minimal simulation sufficient to test them uses the phase-reduced event-gated model

$$d\phi = \omega dt + \sqrt{2D_0} dW_t - \kappa[\phi(t) - \phi(t - \tau_d) - \omega\tau_d]dt, \quad (30)$$

$$\lambda(\phi) = \lambda_0 \exp\left[-\frac{\sin^2 \phi}{2\sigma_\phi^2}\right], \quad (31)$$

$$\dot{I} = -\Gamma I + \sum_j q_j \delta(t - t_j). \quad (32)$$

The subtraction of $\omega\tau_d$ in Eq. (30) removes the trivial phase advance accumulated during the latency interval, so that the delayed term damps phase fluctuations without shifting the nominal clock frequency. Equivalently, one may simulate the fluctuation variable $\psi(t) = \phi(t) - \omega t$, for which $d\psi = \sqrt{2D_0} dW_t - \kappa[\psi(t) - \psi(t - \tau_d)]dt$.

The counted ticks are then extracted by Eq. (8). The simulation should record the raw candidate-event sequence, the analog detector current, and the final counted ticks. The analysis estimates

$$\langle\tau\rangle, \quad \mathcal{N} = \frac{\langle\tau\rangle^2}{\text{Var}(\tau)}, \quad D_\phi = \frac{\Delta\phi\omega}{2\mathcal{N}}, \quad \sigma_A^2(m). \quad (33)$$

As an explicit benchmark, Fig. 2 shows a direct Euler–Maruyama check of the collective scaling laws in the phase-reduced limit $\tau_d = 0$, $\kappa = 0$, $\Gamma \rightarrow \infty$, and one tick per period ($\Delta\phi = 2\pi$). For each M , the simulation uses 2×10^4 independently integrated inter-tick intervals of Eq. (30) with $\omega = 1$, $D_1 = 0.05$, and time step $dt = 3 \times 10^{-3}$. The expected invariants $D_\phi(M)M$ and $\mathcal{N}(M)/M$ are flat within finite-sample error, providing the minimal numerical support requested by Eq. (34).

For the collective test, one repeats Eq. (31) for M independent event channels and drives the same phase variable with the normalized aggregate event stream. The expected pass condition is

$$D_\phi(M)M \simeq \text{const}, \quad \mathcal{N}(M)/M \simeq \text{const}, \quad (34)$$

until finite detector bandwidth or comparator blanking causes missed counts. For the delay test, one sweeps τ_d at fixed κ and checks whether the measured $D_\phi^{\text{eff}}(\tau_d)$ follows the small-signal loop transfer function inferred independently from a weak phase impulse. This benchmark is intentionally modest: it does not prove a universal thermodynamic inequality, but it makes the central claims of Secs. IV–VI falsifiable with a few hundred lines of stochastic simulation code.

VIII. FALSIFIABLE PREDICTIONS

The framework yields the following quantitative tests, each expressible as a scaling law or null test on measurable quantities.

1. **Collective diffusion law.** In the collective regime the product of phase diffusion and unit number is constant, $D_\phi(M)M = \text{const}$, equivalently $\mathcal{N}(M)/M = \text{const}$, until the detector dead time begins to drop ticks. A measured deviation from linearity that is not attributable to dead time falsifies the independence assumption underlying (20) (e.g. would indicate inter-unit correlations such as superradiant enhancement).
2. **Delay-resonant accuracy modulation.** The accuracy $\mathcal{N}(\tau_d)$ is modulated in the loop delay, with extrema fixed by the measured loop transfer function, as in Eqs. (16)–(18). A flat $\mathcal{N}(\tau_d)$ over a range where the loop gain is independently nonzero would falsify the delayed-feedback character of the escapement.

3. **Resolution–accuracy trade-off.** The product $\mathcal{N}\nu$ (and $\mathcal{N}\nu^2$) must respect the bounds of Refs. [5, 6]; exceeding them would indicate a measurement or accounting error rather than new physics.

4. **Extraction-cost balance.** Equation (26) is a directly testable hypothesis: vary the detector bandwidth Γ and threshold I^* and measure whether the inclusion of $\dot{\Sigma}_{\text{det}}$ restores $2\mathcal{N}/[\langle\tau\rangle(\dot{\Sigma}_{\text{engine}} + \dot{\Sigma}_{\text{det}})] \leq 1$. A persistent violation even with the extraction cost included is itself a clean, publishable result.

IX. DISCUSSION

We have shown that a minimal event-gated delayed-feedback oscillator, read out through a finite-bandwidth detector, provides a common description of stochastic-transition, optomechanical-pendulum and delay-feedback clocks, and that within this description two open problems become tractable. First, phase reduction yields the dimensionally consistent accuracy $\mathcal{N} = \Delta\phi\omega/(2D_\phi)$ and a transparent route to the collective scaling $D_\phi(M) \propto M^{-1}$, from which the persistence of TUR violation across the quantum-to-classical transition follows analytically for arbitrary M , beyond the reach of direct simulation. Second, by treating tick extraction as a physical dynamical subsystem with its own entropy production we convert the question “is the violation real?” into a measurable balance, Eq. (26), positioned explicitly relative to recent work on the entropic cost of classical tick extraction [15].

The framework is intentionally agnostic about microscopic realization: the oscillator may be mechanical, electronic, optomechanical or superconducting, and the escapement may be a single emitter, a Josephson element, or a digital controller implementing the latency of Eq. (5). The distinctive structural ingredient—an escapement realized as event-triggered delayed feedback—is what links the otherwise separate literatures of autonomous quantum clocks [2, 14] and delayed nonlinear dynamics [22, 23], and it predicts a delay-dependent accuracy modulation [Eq. (16)] absent from instantaneous-escapement models.

Natural extensions include inter-unit correlations in the collective limit (which would modify Eq. (20) and the scaling (21)), and a full stochastic-thermodynamic treatment of the comparator in Eqs. (6)–(8) to compute $\dot{\Sigma}_{\text{det}}$ from first principles.

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Declarations

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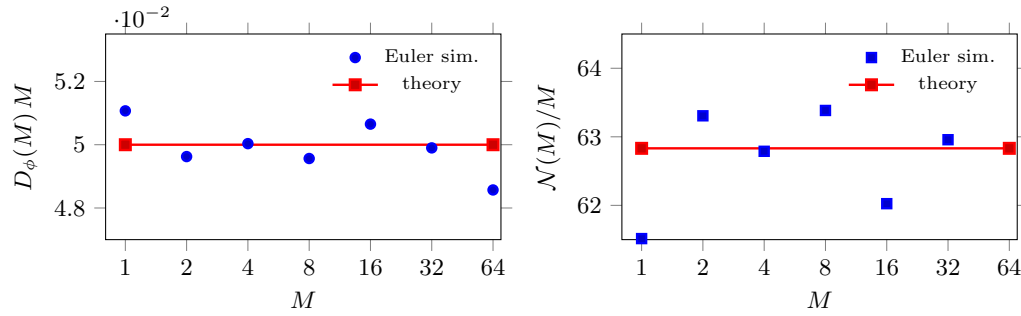


FIG. 2. Minimal numerical benchmark for the collective scaling law in the phase-reduced limit ($\tau_d = 0$, $\kappa = 0$, ideal detector, one tick per period, $\omega = 1$, $D_1 = 0.05$). For each M , 2×10^4 inter-tick intervals were obtained by Euler–Maruyama integration of Eq. (30) with $dt = 3 \times 10^{-3}$. Left: the product $D_\phi(M)M$ is constant within sampling error, consistent with Eq. (20). Right: the normalized accuracy $N(M)/M$ is constant within sampling error, consistent with Eq. (21).

Competing interests

On behalf of all authors, the corresponding author states that there is no conflict of interest.

Consent for publication

Not applicable.

Data availability

No datasets were generated or analyzed during the current study.

Ethics approval

Not applicable.

Code availability

Not applicable.

Consent to participate

Not applicable.

Author contributions

O.I. conceived the study, developed the analysis, performed the derivations, and wrote the manuscript.

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